Essential in Ansys (40 Hours)				
S.No	**Session**	**Topics**	**Type**	**Duration (Hours)**
1	Session 1	Introduction to CAE General procedure to conduct FEA FEA software Key Assumptions in FEA Types of Engineering Analysis Important terms and definitions Classification of materials	Theory/Demo	2.0 Hrs
2	Session 2	Exercises included for practice	Practicals	2.0 Hrs
3	Session 3	**Introduction to ANSYS Workbench** **Sketching and Part Modeling in Design Modeler** Introduction to Modeling Introduction to Design Modeler Window **Illustration 1:** I-section	Theory/Demo	2.0 Hrs
4	Session 4	Exercises included for practice	Practicals	2.0 Hrs
5	Session 5	**Solid Modeling Fundamentals** Overview Introduction Extrusion Sketching	Theory/Demo	2.0 Hrs
6	Session 6	Exercises included for practice	Practicals	2.0 Hrs
7	Session 7	Defining Material Properties Introduction to Engineering Workspace Creating and Adding Materials Assigning Material to the Beam Assigning Material to the Assembly	Theory/Demo	2.0 Hrs
8	Session 8	Exercises included for practice	Practicals	2.0 Hrs
9	Session 9	**Meshing** Introduction Meshing of Plate with Holes	Theory/Demo	2.0 Hrs
10	Session 10	Exercises included for practice	Practicals	2.0 Hrs
11	Session 11	Generating the mesh, optimize the model and generating the local mesh Assembly Meshing	Theory/Demo	2.0 Hrs
12	Session 12	Exercises included for practice	Practicals	2.0 Hrs
13	Session 13	**Static Structural Analysis** Introduction to Static Structural Analysis Pre-processing Solution Post-processing **Static Structural Analysis of:** Cantilever Beam, Plate with a central circular holes, Plate with a square slot, Pressure vessel, Bracket Clevis assembly	Theory/Demo	2.0 Hrs
14	Session 14	Exercises included for practice	Practicals	2.0 Hrs
15	Session 15	**Wizard and Tools** Overview Introduction **Static loadings:** Ductile materials, Brittle materials **Fatigue loading:** Ductile material	Theory/Demo	2.0 Hrs
16	Session 16	Exercises included for practice	Practicals	2.0 Hrs
17	Session 17	**Surface and Line Model** Overview Introduction Sheet with circular hole-plane stress Bracket Line body model	Theory/Demo	2.0 Hrs
18	Session 18	Exercises included for practice	Practicals	2.0 Hrs
19	Session 19	**Natural Frequencies** Overview Introduction Performing the Modal analysis Specifying analysis settings **Modal analysis:** Cantilever beam, Simply supported beam, Chime, Connecting rod, Motor cover Assembly	Theory/Demo	2.0 Hrs
20	Session 20	Exercises included for practice	Practicals	2.0 Hrs